

Xiaomeng Yang

(+1)-857-565-5472 | yang.xiaome@northeastern.edu | xiaomeng-yang.github.io

EDUCATION

Northeastern University

Ph.D. Computer Engineering

Boston, MA

Sep. 2024 – June 2029 (Expected)

University of Chinese Academy of Sciences (UCAS)

M.S. Computer Application Technology

Beijing, China

Sep. 2021 – June 2024

Zhejiang University (ZJU)

B.E. Electronic and Computer Engineering

Zhejiang, China

Sep. 2017 – June 2021

University of Illinois at Urbana-Champaign (UIUC)

B.S. Computer Engineering with Honors

Champaign, IL

Sep. 2017 – May 2021

RESEARCH INTERESTS

Video Generation and Diffusion Models; World Models (video-based simulation)

Efficient Generative Models, Model Compression and Acceleration

Multimodal Learning, OCR, Scene Text Recognition, TextVQA

PUBLICATIONS

- **Xiaomeng Yang**, Yanyu Li, Gordon Guocheng Qian, Ivan Skorokhodov, Viacheslav Ivanov, Avalon Vinella, Xuan Zhang, Yanzhi Wang, Sergey Tulyakov, Anil Kag. Prompt2Effect: Training-Free Image-to-Video Model Specialization via LoRA Generation. In ECCV, 2026
- **Xiaomeng Yang***, Lei Lu*, Qihui Fan, Changdi Yang, Juyi Lin, Yanzhi Wang, Xuan Zhang, Shangqian Gao. ALTER: All-in-One Layer Pruning and Temporal Expert Routing for Efficient Diffusion Generation. In NeurIPS, 2025.
- **Xiaomeng Yang**, Jian Gao, Yanzhi Wang, Xuan Zhang. ZEROSIM: Zero-Shot Analog Circuit Evaluation with Unified Transformer Embeddings. In ICCAD, 2025
- **Xiaomeng Yang**, Zhi Qiao, Yu Zhou. IPAD: A Parallel, Iterative and Diffusion-Based Network for Scene Text Recognition. In International Journal of Computer Vision (IJCV), 2025.
- Yifei Zhang, Chang Liu, Jin Wei, **Xiaomeng Yang**, Yu Zhou, Can Ma, Xiangyang Ji. Linguistics-aware Masked Image Modeling for Self-supervised Scene Text Recognition. Accepted by CVPR, 2025.
- **Xiaomeng Yang**, Zhi Qiao, Jin Wei, Dongbao Yang, Yu Zhou. Masked and Permuted Implicit Context Learning for Scene Text Recognition. In IEEE Signal Processing Letters, 2024.
- **Xiaomeng Yang**, Dongbao Yang, Zhi Qiao, Yu Zhou. Accurate and Robust Scene Text Recognition via Adversarial Training. In ICASSP, 2024.
- Gangyan Zeng, Yuan Zhang, Yu Zhou, **Xiaomeng Yang**, Ning Jiang, Guoqing Zhao, Weiping Wang, Xu-Cheng Yin. Beyond OCR+VQA: Towards End-to-End Reading and Reasoning for Robust and Accurate TextVQA. In Pattern Recognition, Volume 138, 2023.
- Gangyan Zeng, Yuan Zhang, Yu Zhou, **Xiaomeng Yang**. Beyond OCR+VQA: Involving OCR into the Flow for Robust and Accurate TextVQA. In ACM MM, 2021.
- Gangyan Zeng, Yuan Zhang, Yu Zhou, **Xiaomeng Yang**. A Cost-Efficient Framework for Scene Text Detection in the Wild. In PRICAI, 2021.
- Yucheng Jin, **Xiaomeng Yang**, Chengting Yu, Liangjing Yang. Educational Data Mining: Discovering Principal Factors for Better Academic Performance. In BDET, 2021.

RESEARCH EXPERIENCE

Weight-Driven LoRA Synthesis for Efficient VFX Adaptation

Sep. 2025 – Mar. 2026

Snap Inc. | Internship

- Designed *Prompt2Effect*, a weight-driven hypernetwork that synthesizes effect-specific LoRA weights in a single forward pass, eliminating the need for costly per-effect LoRA fine-tuning in Image-to-Video generation.
- Engineered an SVD-canonicalized parameterization explicitly conditioned on frozen base model weights, resolving factorization ambiguity and stabilizing high-dimensional weight synthesis.
- Achieved a reduction in processing time from 56 GPU training hours to 3.3 seconds and a 10x acceleration in downstream fine-tuning, all while maintaining on-par or superior video quality.

Diffusion Models Acceleration Based on Dynamic Model Sparsity

Nov. 2024 – May. 2025

NEU | *Research Assistant, Advisor: Prof. Yanzhi Wang*

- Investigated the temporal Mixture-of-Experts (MoE) architecture for diffusion models, enabling expert specialization across different denoising timesteps to reduce inference redundancy.
- Implemented an integrated all-in-one pruning and routing strategy that jointly learns sparse substructures and expert selection via a trainable timestep-aware decision generator. This work is accepted by NeurIPS 2025.

AI for Analog Circuit Evaluation

Nov. 2024 – Apr. 2025

NEU | *Research Assistant, Advisor: Prof. Xuan Zhang*

- Explored machine learning methods for efficient evaluation of analog circuits. Aim to accelerate the simulation process using transformer or other advanced neural networks instead of traditional simulation.
- Proposed ZeroSim, a unified transformer-based network to perform the Amplifier evaluation. It shows high accuracy for both in-distribution and out-of-distribution Amplifier performance evaluation. This work led to a paper accepted by ICCAD 2025.
- Extending the evaluation model to more circuit types to make it powerful across different kinds of circuits.

Linguistic Information Enhanced Scene Text Recognition

June 2022 – Apr. 2024

UCAS | *Research Assistant, Advisor: Prof. Yu Zhou*

Tomorrow Advancing Life | *Internship, Mentor: Zhi Qiao*

- Explored the relationship between the visual features and text information in scene text recognition.
- Proposed a framework for autoregressive decoding using masked and permuted implicit context learning under additional length supervision. Enhanced model robustness against length prediction errors through perturbation training. Resulted in a paper in IEEE Signal Processing Letters 2024.
- Explored non-autoregressive decoding in scene text recognition as image-based conditional text generation. Utilized discrete diffusion with a small transformer decoder and easy-first decoding for enhanced bidirectional context awareness. The manuscript is accepted by IJCV 2025.

Adversarial Training for Scene Text Recognition

Mar. 2023 – Aug. 2023

UCAS | *Summer Internship, Advisor: Prof. Yu Zhou*

- Investigated the impact of adversarial training on various scene text recognition models.
- Implemented adversarial attacks on extensively used CTC-based and attention-based methods. Utilized adversarial training as a model-dependent augmentation with a regularization norm. The model gained robustness and improved performance, especially on challenging images. This work led to a paper accepted by ICASSP 2024.

Involving OCR into the Flow of TextVQA

Oct. 2020 – Sep. 2022

UCAS | *Research Assistant, Advisor: Prof. Yu Zhou*

- Developed two accurate TextVQA models by integrating OCR into the process.
- Designed a reading modification module for adaptive answer selection based on context. Visually enhanced text embedding and semantically oriented object embedding were used to enhance the feature representations. This approach is more robust to the OCR errors, leading to a paper published by ACM MM 2021.
- Devised an efficient end-to-end network integrating text reading and reasoning, where the downstream VQA signal contributes to the optimization of text reading. The approach outperformed existing alternatives in accuracy and robustness, resulting in a publication in Pattern Recognition 2023.

SKILLS

Language skills: English: Advanced, Chinese: Native

Programming languages: Python, C/C++, Java, JavaScript, HTML/CSS

Other Library and Tools: PyTorch, TensorFlow, Numpy, Scipy, Linux, Git